

Supplementary Material: State and Action Space for Imitation Learning in Pediatric ECMO

1 Overview

This supplementary note provides the detailed state and action definitions used in our imitation-learning framework for pediatric extracorporeal membrane oxygenation (ECMO). In the main paper, we summarize the representation briefly due to space constraints. Here, we describe how hourly patient trajectories are converted into state-action pairs for supervised policy learning.

Each patient trajectory is represented as a sequence of tuples

$$\tau^{(i)} = \{(s_t^{(i)}, a_t^{(i)})\}_{t=1}^{T_i}, \quad (1)$$

where $s_t^{(i)}$ is the clinical state of patient i at hour t , and $a_t^{(i)}$ is the inferred clinician action vector at that time. For readability, we omit the patient superscript in the remainder of this document.

2 Temporal Aggregation

The raw ECMO record contains measurements from multiple clinical data streams, including hemodynamics, ECMO circuit configuration, ventilator readings, and laboratory tests. Because measurements are not always recorded at identical times or frequencies, we discretize each trajectory into non-overlapping 60-minute windows. For each variable, the value at time t is computed as the mean of all available measurements within the corresponding hourly window. This produces a regularly sampled trajectory suitable for sequential policy learning.

3 State Space

At each time step t , the state vector s_t is constructed from 23 physician-selected physiologic variables. For each variable, we include both its current hourly value and its one-hour change relative to the previous window. Age-sensitive variables, heart rate (HR) and mean arterial pressure (ARTm), are normalized according to the patient’s age. In addition, we include the type of ECMO support and a binary indicator denoting whether the patient is on ECMO at the given time step.

Thus, the final state representation contains

$$23 \text{ current values} + 23 \text{ one-hour deltas} + 1 \text{ ECMO type feature} + 1 \text{ on-ECMO indicator} = 48 \quad (2)$$

features.

Let z_t^j denote the hourly value of physiologic variable j at time t , where $j \in \{1, \dots, 23\}$. The one-hour delta feature is defined as

$$\Delta z_t^j = z_t^j - z_{t-1}^j. \quad (3)$$

The state vector can then be written as

$$s_t = [z_t^1, \dots, z_t^{23}, \Delta z_t^1, \dots, \Delta z_t^{23}, e_t, o_t], \quad (4)$$

where e_t represents ECMO type and $o_t \in \{0, 1\}$ indicates whether the patient is on ECMO.

Table 1: Physiologically curated variables used to construct the state vector. For each variable, both the current hourly value and the one-hour delta are included. HR and ARTm are age-normalized.

Category	Feature name	Unit
Hemodynamics	Mean Arterial Pressure (ARTm)	mmHg
	Heart Rate (HR)	bpm
	SpO ₂	%
	Cerebral Oximetry (rSO ₂ -1, rSO ₂ -2)	%
ECMO Circuit	Blood Flow	mL/kg/min
	Sweep Gas: CO ₂ Flow	L/min
	Sweep Gas: O ₂ Flow	L/min
	FiO ₂ – ECMO	%
	Volume Sensor	mmHg
Ventilator	Mean Airway Pressure (P_{AW})	cmH ₂ O
	Positive End-Expiratory Pressure (PEEP)	cmH ₂ O
	Oxygen Concentration (FiO ₂)	%
	Tidal Volume	mL/kg
	End-tidal Carbon Dioxide (etCO ₂)	mmHg
Laboratory	Arterial Partial Pressure of Oxygen (PO ₂)	mmHg
	Arterial Partial Pressure of Carbon Dioxide (PCO ₂)	mmHg
	Arterial pH	–
	Arterial Base Excess	mmol/L
	Lactate	mmol/L
	Ionized Calcium	mmol/L
	Total CO ₂	mEq/L
	International Normalized Ratio (INR)	–

Additional state indicators. In addition to the 23 physiologic variables and their deltas, the state includes ECMO type and an on-ECMO indicator. ECMO type captures the mode of support, such as veno-venous (VV) or veno-arterial (VA) ECMO. The on-ECMO indicator is a binary variable that denotes whether the patient is actively receiving ECMO support at the corresponding time point.

4 Action Space

The raw clinical record does not contain explicit, timestamped labels for clinician interventions. Therefore, we infer action labels from clinically meaningful changes in selected actionable variables. We use four actionable features: arterial PO₂, arterial PCO₂, SpO₂, and FiO₂ – ECMO. Each actionable feature is treated as an independent action head.

For each actionable feature k , the inferred action at time t is a three-class label:

$$a_t^k \in \{\text{Same}, \text{Decrease}, \text{Increase}\}. \quad (5)$$

The label is determined by comparing the change in the actionable feature over the next 60-minute interval to a physician-defined threshold δ_k :

$$a_t^k = \begin{cases} \text{Increase,} & x_{t+1}^k - x_t^k > \delta_k, \\ \text{Decrease,} & x_{t+1}^k - x_t^k < -\delta_k, \\ \text{Same,} & \text{otherwise.} \end{cases} \quad (6)$$

Here, x_t^k is the hourly value of actionable feature k at time t . The thresholds are designed to suppress small physiologic fluctuations and measurement noise, so that the inferred labels reflect clinically meaningful changes.

Table 2: Actionable features and physician-defined thresholds used to infer discrete action labels. A change is identified when the absolute change over a 60-minute window exceeds the threshold.

Action head	Threshold	Unit
Arterial PO ₂	25	mmHg
Arterial PCO ₂	5	mmHg
SpO ₂	5	%
FiO ₂ – ECMO	10	%

The complete action vector at time t is therefore

$$a_t = (a_t^1, a_t^2, a_t^3, a_t^4), \quad (7)$$

where each component corresponds to one of the four action heads. Since each head has three possible labels, the full joint action space has size

$$3^4 = 81. \quad (8)$$

However, directly modeling all 81 joint actions would increase sparsity, especially because many combinations are rare or absent in the pediatric ECMO data. Instead, we learn one predictor per action head, allowing simultaneous multi-variable recommendations while avoiding the combinatorial expansion of the joint action space.

5 Hierarchical Action Labeling

Because most time steps correspond to no clinically meaningful change, the action labels are highly imbalanced toward the “Same” class. To address this imbalance, each action head is represented using a two-stage hierarchy.

First, we define a binary change indicator:

$$c_t^k = \begin{cases} 1, & a_t^k \in \{\text{Decrease, Increase}\}, \\ 0, & a_t^k = \text{Same.} \end{cases} \quad (9)$$

Second, conditional on a change occurring, we define a direction label:

$$d_t^k \in \{\text{Decrease, Increase}\}. \quad (10)$$

At inference time, the predicted binary change decision and direction decision are merged into the final three-class label:

$$\hat{y}_t^k = \begin{cases} \text{Same,} & \hat{c}_t^k = 0, \\ \text{Decrease,} & \hat{c}_t^k = 1 \text{ and } \hat{d}_t^k = \text{Decrease}, \\ \text{Increase,} & \hat{c}_t^k = 1 \text{ and } \hat{d}_t^k = \text{Increase}. \end{cases} \quad (11)$$

This decomposition encourages the model to first identify whether a clinically meaningful change is likely, and then determine the direction of that change only when needed.

6 Policy-Learning Formulation

Under a first-order Markov assumption, the constructed state s_t is treated as a sufficient summary of the relevant clinical history. The expert clinician action distribution is therefore approximated as

$$p(A_t \mid s_{1:t}) \approx p(A_t \mid s_t). \quad (12)$$

Rather than estimating the full joint distribution over all action heads, we estimate head-specific policies:

$$\pi_k(a^k \mid s_t) \approx p(A_t^k = a^k \mid s_t), \quad k = 1, \dots, 4. \quad (13)$$

The learned multi-head policy is then represented as

$$\pi(a_t \mid s_t) = \left(\pi_1(a_t^1 \mid s_t), \dots, \pi_4(a_t^4 \mid s_t) \right), \quad (14)$$

which permits multiple actionable features to be adjusted at the same time while keeping each prediction problem statistically tractable.

7 Summary

In summary, each hourly ECMO observation is represented by a 48-dimensional state vector composed of 23 physiologic variables, their one-hour deltas, ECMO type, and an on-ECMO indicator. Clinician actions are inferred for four actionable features using physician-defined thresholds over 60-minute windows. Each action head is modeled as a three-class decision and implemented using a two-stage hierarchy for change detection and direction prediction. This representation converts observational ECMO trajectories into supervised demonstrations for learning clinician-behavior policies.